Tristan Tan

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Education

University of California, Berkeley — Berkeley, CA
B.A. Data Science and Economics; GPA: 3.88 — *Expected May 2029*

Coursework: Data Science (Foundations, Comp. Structures), Probability & Statistics, Linear Algebra, Microeconomics

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, R, HTML/CSS
Data & ML: pandas, NumPy, SciPy, scikit-learn, PyTorch, Sentence Transformers, OpenCV, Matplotlib, Tableau
Frameworks & Tools: React, Next.js, FastAPI, PostgreSQL, Supabase, MongoDB, Git, Vercel, OAuth 2.0, REST APIs

Experience

Undergraduate Research Assistant — Sep 2026 – Present
Archaeological Research Facility, UC Berkeley — *Berkeley, CA*

- Developed an interactive 3D annotation platform (Next.js, TypeScript, React Three Fiber) mapping 314 scholarly annotations to 607 spatial regions across 6 photogrammetric Egyptian coffin models
- Implemented 3D mesh raycasting and bidirectional spatial selection, letting users navigate between hieroglyphic inscriptions, annotated surfaces, and their scholarly translations
- Built a data pipeline linking annotation records to 3D geometry, replacing manual XML-based lookup

Data Research Intern — Sep 2026 – Present
Ten Strands — *Berkeley, CA*

- Analyzed climate preparedness in 937 California school districts across 200+ demographic and policy indicators
- Identified district enrollment as the strongest predictor of documented climate protections in county-clustered regression; the urban–rural gap disappeared after controlling for district size
- Identified 199 high-wildfire-risk and 176 high-pollution districts with no matching documented protections, and mapped geographic clusters with spatial autocorrelation (Moran's I) analysis

Data Science Intern — May 2026 – Aug 2026
IDX Exchange — *Remote*

- Built home valuation models on 297K+ California sales over 27 months, engineering property and market features
- Compared Linear Regression, Random Forest, and XGBoost with chronological backtesting on 11K+ held-out sales, reaching 0.92 R^2, 6% median error, and 34% lower MAE than the linear baseline

Undergraduate Machine Learning Researcher — Aug 2025 – May 2026
Lucarelli Photogrammetry Lab, UC Berkeley — *Berkeley, CA*

- Built PyTorch PointNet and multi-head self-attention models predicting camera-coverage gaps from camera layouts
- Benchmarked against Ridge regression across five-seed synthetic experiments, achieving 42–49% lower MAE with attention; built a camera-guided ICP evaluation to measure geometric reconstruction error

Leadership

Marketing Officer — Aug 2025 – May 2026
Berkeley ANova — *Berkeley, CA*

- Taught weekly Python lessons and programming labs to 100+ students across 7 Bay Area schools
- Led digital marketing and outreach campaigns generating 100,000+ impressions and increasing recruitment engagement by 137% through data-driven content strategy and cross-campus partnerships

Lead Product Designer — May 2025 – Jul 2025
IxD @ ArtCenter College of Design — *Pasadena, CA*

- Led end-to-end UX/UI design of Pasadena High School's App Academy website with ArtCenter designers, producing high-fidelity Figma prototypes, visual branding, and responsive interfaces
- Ran user research and usability testing (card sorting, focus groups, A/B walkthroughs) and turned stakeholder feedback into a launched site built with HTML, CSS, JavaScript, and Vue.js